

# Projective Information Loss and Fibre Erasure in Admissible Non-Injective Projection

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## Abstract

We establish a fibre-erasure theorem for the admissible coarse-graining hierarchy of the Cosmochrony spectral programme [3, 1]. The non-injective projection  $\Pi : \Omega \rightarrow \mathcal{O}$  defines a residual fibre-information functional  $I(c; \sigma(\ell))$ , where  $c$  is a Weil-block fibre label and  $\sigma_c(\ell)$  is the BFS capacity profile at coarse-graining depth  $\ell$ . Conditional on the structural sufficiency hypothesis [H-suff] — that  $\sigma(\ell)$  is a sufficient statistic for the hierarchy beyond  $\ell$ , a condition numerically supported for  $q \in \{61, 151, 211\}$  (Section 5.3) — the data-processing inequality [9] implies that  $I(c; \sigma(\ell))$  is non-increasing in  $\ell$ : fibre information is erased, not created, under admissible coarse-graining. This result is *not* a  $c$ -theorem; it does not count effective degrees of freedom. It is a quantitative realisation of the ENI no-go theorem [3]: non-injectivity of  $\Pi$  forces projective information loss, and the data-processing inequality makes this loss monotone and measurable. The inter-sector variance  $\text{Var}_c(\sigma_c(\ell))$  serves as a computable proxy; its restriction to  $\text{SU}(3)$  colour-triplets connects directly to hypothesis [H-color] [8, 7].

**Keywords.** non-injective projection; fibre erasure; admissible coarse-graining; data-processing inequality; information monotone; BFS capacity; spectral admissibility; projective information loss; Weil representation; Heisenberg group; [H-color]

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# 1 Introduction

The ENI no-go theorem [3] establishes a fundamental structural fact: genuine emergence is necessarily non-injective. Any surjective map  $\Pi : \Omega \rightarrow \mathcal{O}$  satisfying the projective-emergent conditions carries a strictly positive projection entropy  $S_\Pi > 0$ : information about the microscopic representative is irrecoverably lost at the effective level. This is a qualitative result. It identifies the structure of information loss but says nothing about how this loss is distributed across a hierarchy of resolutions.

The Cosmochrony spectral programme [5, 6] provides a concrete realisation of this hierarchy. For each prime  $q$ , the BFS capacity profiles  $\sigma_c(\ell)$ , indexed by Weil-block fibre label  $c$  and coarse-graining depth  $\ell$ , track the residual projective capacity of block  $c$  as the BFS cascade advances. Numerical campaigns across  $q \in \{61, 151, 211, 307\}$  reveal a consistent pattern: inter-sector variance collapses as depth increases, and the profiles converge toward a universal profile  $\sigma^*$  approached in the limit  $q \rightarrow \infty$  (Section 5.4). Hypothesis [H-color] [8, 7] identifies one specific instance of this convergence: the  $SU(3)$  colour-triplet sectors  $\{c, \omega c, \omega^2 c\}$  become indistinguishable at the effective level for  $q \equiv 1 \pmod{3}$ .

This paper addresses the following question: can the qualitative ENI no-go and the quantitative spectral convergence be connected by a principled information-theoretic structure? We show that the answer is yes, conditional on the structural sufficiency hypothesis [H-suff]: the residual fibre-information functional  $I(c; \sigma(\ell))$  is non-increasing in the coarse-graining depth  $\ell$ , by the data-processing inequality [9]. The result is conditional, not unconditional: [H-suff] is strongly motivated (Remark 3.2), numerically supported for  $q \in \{61, 151, 211\}$  (Section 5.3), but not yet proved analytically.

Two structural distinctions are essential before proceeding. First, the coarse-graining axis  $\ell$  is distinct from and orthogonal to the temporal cascade axis  $n$  of the PTO paper [4]: see Table 1 and Remark 2.3. Second, as stated in Section 1.1, ENI alone does not imply monotonicity; the hierarchical structure of [H-suff] and [H-sg] is the informational content this paper contributes beyond ENI.

## 1.1 Contributions and non-claims

This paper establishes, conditional on hypotheses [H-suff] and [H-sg] (Section 2.4):

- (i) a monotone information functional  $I(c; \sigma(\ell))$  for the BFS coarse-graining hierarchy (Theorem 4.3);
- (ii) its non-increase under depth-increment, via the data-processing inequality;
- (iii) the identification of  $I(c; \sigma(\ell)) = 0$  as the fibre-erasure equilibrium condition (Corollary 4.4).

This paper does *not* establish:

- (i) a  $c$ -theorem or analogue (no counting of effective degrees of freedom; see open problem [O-2]);
- (ii) hypothesis [H-suff] itself (stated and motivated; numerically supported for  $q \in \{61, 151, 211\}$ ; analytical proof remains open, see [O-1]);
- (iii) the semigroup hypothesis [H-sg] (stated and motivated; see open problem [O-1]).

**Strict disclaimer.** This result should not be called a  $c$ -theorem. It is a *fibre-erasure theorem*: the monotone measures the loss of residual fibre-label information under admissible coarse-graining. A  $c$ -theorem-type statement would require an additional counting principle for effective degrees of freedom, which is not established here and is listed as open problem [O-2] in Section 6.

**ENI versus monotonicity.** ENI [3] establishes that information is structurally lost under non-injective projection, but does not by itself imply any monotone hierarchy over the coarse-graining scale  $\ell$ . The monotonicity of Theorem 4.3 requires the additional hierarchical structure encoded in [H-suff] and [H-sg]: these are the informational content this paper contributes beyond ENI.

## 1.2 Programme placement

The present paper occupies a hub position in the programme. ENI [3] and Foundation [1] provide the foundational input: the non-injective projection  $\Pi$ , the Weil-fibre realisation, and the projection entropy  $S_\Pi$ . The spectral admissibility programme [5, 6] provides the operational hierarchy: the BFS capacity profiles  $\sigma_c(\ell)$  and their numerical campaigns, including the O31–O32 investigation [8, 7]. This paper connects these two branches by translating the ENI projection entropy into the operational functional  $I(c; \sigma(\ell))$  and identifying spectral convergence  $\sigma_c \rightarrow \sigma^*$  as the thermodynamic universal profile (Section 5.4). The PTO paper [4] handles the complementary temporal axis  $n$  (Table 1); the two papers are structurally orthogonal and together constitute the information-theoretic architecture of the admissibility cascade.

## 2 Framework

### 2.1 Non-injective projection and fibre structure

We recall the setting of [3, 1].

**Definition 2.1** (Non-injective projection and fibre). A *projective description* is a surjective map  $\Pi : \Omega \rightarrow \mathcal{O}$  such that  $\mathcal{O}$  carries no microstate-resolving structure independent of  $\Pi$ . The *fibre* over  $o \in \mathcal{O}$  is  $\Pi^{-1}(o) = \{\omega \in \Omega : \Pi(\omega) = o\}$ . The *projection entropy* is  $S_\Pi = \mathbb{E}_o[\log |\Pi^{-1}(o)|] \geq 0$ , with  $S_\Pi > 0$  whenever  $\Pi$  is non-injective (ENI Theorem 2.1, [3]).

In the spectral realisation [1], the fibre structure is implemented via the Weil representation  $V_\rho$  of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$ . The minimal non-injectivity is the parity involution  $c \leftrightarrow q - c$ : Weil blocks  $\rho_c$  and  $\rho_{q-c}$  map to the same effective observable class. The *fibre label*  $c \in \{1, \dots, (q-1)/2\}$  indexes blocks within an equivalence class under  $\Pi$ .

### 2.2 The BFS coarse-graining hierarchy

**Definition 2.2** (BFS capacity profile). For a prime  $q$ , a fibre label  $c$ , and BFS depth  $\ell \in \mathbb{N}$ , the *BFS capacity profile*  $\sigma_c(\ell) \in [0, 1]$  is the normalised residual projective capacity of block  $c$  at depth  $\ell$  along the Heisenberg Cayley graph BFS cascade [5, 6]. The *pair observable* is  $\sigma_{\text{pair}}(\ell) = \sigma_c(\ell) \cdot \sigma_{q-c}(\ell)$  on conjugate pair  $(c, q - c)$ .

For a prime  $q$ , the Heisenberg group  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  carries a Cayley graph whose BFS shells  $B_\ell$  grow as  $|B_\ell| \sim \ell^4$  in the Heisenberg regime [5]. The capacity profile  $\sigma_c(\ell)$  measures the residual Gram-Schmidt span of Weil block  $\rho_c$  not yet exhausted at shell depth  $\ell$ : it starts at  $\sigma_c(0) = 1$  and decreases toward 0 as BFS saturation is approached. The depth parameter  $\ell \in \mathbb{N}$  is discrete; the coarse-graining semigroup of Section 2.4 therefore acts on a discrete monoid, not a continuous scale.

### 2.3 Three-axis separation

*Remark 2.3* (Cascade and coarse-graining are orthogonal). The admissible cascade  $U_0 \rightarrow U_1 \rightarrow \dots$  defines the temporal axis  $n$  and is *not* an RG flow. RG-like behaviour is carried by  $\Pi$  and the semigroup  $\{G_{\ell \leftarrow \ell}\}$  on the  $\ell$ -axis. The cascade *enlarges* the admissible space as temporal ordering advances; an RG flow toward the IR *reduces* effective degrees of freedom. These are distinct

| Axis                         | Parameter                 | Role                                                                 |
|------------------------------|---------------------------|----------------------------------------------------------------------|
| Coarse-graining / resolution | $\ell$ (BFS depth/window) | coarse-graining hierarchy; carries $I(c; \sigma(\ell))$ ; this paper |
| Thermodynamic / continuum    | $q \rightarrow \infty$    | Limit for $\sigma^*$ ; established numerically (Section 5.4)         |
| Temporal / cascade           | $n$ (cascade step)        | Projective time; carries $I(n)$ of [4]                               |

Table 1: The three distinct axes of the observable hierarchy. The coarse-graining axis  $\ell$  is the support of this paper. The temporal axis  $n$  is the support of PTO [4]. The continuum axis  $q \rightarrow \infty$  is the thermodynamic limit in which the fixed point  $\sigma^*$  is approached; Section 5.4 establishes this numerically.

phenomena on orthogonal axes. The fibre-erasure fixed point  $\sigma^*$  lives on the thermodynamic axis: it is approached as  $q \rightarrow \infty$  with  $O(q^{-1/2})$  corrections, not as a Wilsonian  $\ell$ -limit at fixed  $q$  (Section 5.4).

## 2.4 The coarse-graining semigroup

**Definition 2.4** (Coarse-graining channel). For  $\ell' > \ell$ , the *coarse-graining channel*  $G_{\ell' \leftarrow \ell} : \sigma(\ell) \mapsto \sigma(\ell')$  is the map sending the BFS capacity profile at depth  $\ell$  to the profile at depth  $\ell'$ .

**Hypothesis 2.5** ([H-sg]: semigroup property). The coarse-graining channels satisfy

$$G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1} \quad \text{for all } \ell_1 < \ell_2 < \ell_3. \quad (1)$$

Hypothesis [H-sg] is plausible from the nested structure of BFS shells: the subgraph induced by  $B_{\ell_1}$  is contained in that induced by  $B_{\ell_2}$  for  $\ell_1 < \ell_2$ , so the transition  $\sigma(\ell_1) \rightarrow \sigma(\ell_2)$  can in principle be factored through  $\sigma(\ell_1)$  alone. Without [H-sg], the data-processing inequality applied step-by-step via [H-suff] yields  $I(c; \sigma(\ell+1)) \leq I(c; \sigma(\ell))$  for each consecutive pair, but not the uniform statement of Theorem 4.3 for all  $\ell' > \ell$ . Note that the semigroup acts on the space of observable descriptions, not on the static substrate  $\Omega$ : composability is a property of the coarse-graining of effective observables.

## 2.5 Probability measure on fibre labels

**Definition 2.6** (Fibre-label measure). Let  $\mu$  be a probability measure on  $\{c : 1 \leq c \leq (q-1)/2\}$ . Two canonical choices are used:

- (i) *Uniform*:  $\mu(c) = 2/(q-1)$  for all  $c$ .
- (ii) *Triplet-uniform* (for  $q \equiv 1 \pmod{3}$ ):  $\mu$  uniform on  $\{c, \omega c, \omega^2 c\}$ , where  $\omega$  is a primitive cube root of unity modulo  $q$ ; used in the [H-color] context [8, 7].

The choice of  $\mu$  does not affect the monotonicity of  $I(c; \sigma(\ell))$  (which holds for any fixed measure by DPI), but affects its value and the interpretation of the fixed point.

## 3 Structural Sufficiency and the Markov Property

**Hypothesis 3.1** ([H-suff]: structural sufficiency). For all  $\ell' > \ell$ , the coarse-graining channel  $G_{\ell' \leftarrow \ell}$  is conditionally independent of the fibre label  $c$  given  $\sigma(\ell)$ :

$$c \longrightarrow \sigma(\ell) \longrightarrow \sigma(\ell'). \quad (2)$$

That is,  $\sigma(\ell)$  is a *sufficient statistic* for  $\sigma(\ell')$  with respect to the fibre label  $c$ . The term “sufficient” is used in the information-theoretic sense: all information in  $\sigma(\ell)$  that is predictive of  $\sigma(\ell')$  is already fully encoded; no additional knowledge of  $c$  reduces the uncertainty in  $\sigma(\ell')$  beyond what  $\sigma(\ell)$  already captures.

*Remark 3.2* (Interpretive status of [H-suff]). Hypothesis [H-suff] is not introduced solely to enable the DPI. It is the informational reformulation of an effective closure property already implicit throughout the spectral admissibility programme: if [H-suff] failed — if BFS dynamics at depth  $\ell'$  re-accessed label information not encoded in  $\sigma(\ell)$  — then  $\sigma_c(\ell)$  would not constitute stable effective observables under  $\Pi$ , and the convergence  $\sigma_c \rightarrow \sigma^*$ , the plateau structure across  $q \in \{61, 151, 211, 307\}$ , and the [H-color] campaign [8, 7] would be structurally very difficult to account for. More precisely, [H-suff] is the condition that the BFS observable  $\sigma(\ell)$  constitutes a *complete summary* of the fibre structure at resolution  $\ell$ : the projection  $\Pi$  has already collapsed the relevant fibre information at each level of the hierarchy, and no deeper BFS step retrieves it. This is exactly what the non-injectivity of  $\Pi$  (ENI [3]) demands at the effective level: fibre information is not gradually re-injected as coarse-graining proceeds, but is collapsed once and for all by  $\Pi$ . In this sense, [H-suff] makes explicit, in information-theoretic language, a structural commitment already implicit in the programme's use of  $\sigma_c(\ell)$  as an effective observable.

*Remark 3.3* (Status of [H-suff]). Hypothesis [H-suff] is strongly motivated by Remark 3.2 and is compatible with all known spectral results. A permutation-based Spearman test on BFS capacity data for  $q \in \{61, 151, 211\}$  yields no significant  $c$ -dependence in step-by-step residuals at any tested depth (Section 5.3), providing numerical evidence for [H-suff]. Analytical proof remains open; see open problem [O-1].

*Remark 3.4* (Why [H-suff] is load-bearing). If hypothesis [H-suff] fails for some pair  $(\ell, \ell')$  — that is, if  $\sigma(\ell')$  depends on  $c$  beyond its dependence on  $\sigma(\ell)$  — then no structural information monotone of the form  $I(c; \sigma(\ell')) \leq I(c; \sigma(\ell))$  can be guaranteed. If  $p(\sigma(\ell') \mid \sigma(\ell), c) \neq p(\sigma(\ell') \mid \sigma(\ell))$  for some  $c$ , then the chain  $c \rightarrow \sigma(\ell) \rightarrow \sigma(\ell')$  is not Markov, and the data-processing inequality does not apply. This is not a limitation of the proof technique; it reflects the genuine logical dependence of the monotonicity on the Markov structure.

A proof of [H-suff] would establish that the BFS depth-increment kernel  $p(\sigma(\ell+1) \mid \sigma(\ell), c)$  is independent of the fibre label  $c$ . Structurally, this would follow from the fact that the BFS recursion on the Heisenberg Cayley graph is governed by the group geometry — shared across all fibre labels up to conjugation — and the Born-Infeld admissibility constraint, rather than by the specific character of block  $c$ . A partial result is already in hand: the conjugation symmetry  $\rho_c \cong \rho_{q-c}$  ensures  $\sigma_c(\ell) = \sigma_{q-c}(\ell)$  within each conjugate pair, consistent with [H-suff] at the pair level. The full extension requires showing that cross-pair dynamics also factors through  $\sigma(\ell)$ .

The full extent of [H-suff] — including cross-pair dynamics — has been tested numerically for  $q \in \{61, 151, 211\}$ ; see Section 5.3.

## 4 The Fibre-Erasure Theorem

### 4.1 The information functional

**Definition 4.1** (Residual fibre information). The *residual fibre information* at depth  $\ell$  is

$$I(c; \sigma(\ell)) = H(c) - H(c \mid \sigma(\ell)) = H(\sigma(\ell)) - H(\sigma(\ell) \mid c), \quad (3)$$

where  $H(\cdot)$  denotes Shannon entropy with respect to measure  $\mu$ .

*Remark 4.2* (Physical interpretation).  $I(c; \sigma(\ell)) = 0$  if and only if  $\sigma(\ell) \perp c$ : the BFS profile carries no information about the microscopic fibre representative. This is the operational statement of fibre erasure under  $\Pi$ .

### 4.2 Main theorem

**Theorem 4.3** (Fibre-Erasure Theorem). *Assume hypotheses [H-suff] (Hypothesis 3.1) and [H-sg] (Hypothesis 2.5). Then for all  $\ell' > \ell \geq 0$ :*

$$I(c; \sigma(\ell')) \leq I(c; \sigma(\ell)). \quad (4)$$

In particular,  $I(c; \sigma(\ell))$  is non-increasing in  $\ell$  and converges to a limit  $I_\infty \geq 0$ .

*Proof sketch.* By [H-suff], the triple  $(c, \sigma(\ell), \sigma(\ell'))$  forms a Markov chain (2). The data-processing inequality [9] applied to this chain gives (4) directly. Composability [H-sg] extends the inequality from  $\ell' = \ell + 1$  to all  $\ell' > \ell$ .  $\square$

**Corollary 4.4** (Fibre-erasure equilibrium condition). *Under the hypotheses of Theorem 4.3,  $I(c; \sigma(\ell)) = 0$  if and only if  $\sigma(\ell) \perp c$ . This is the fibre-erasure equilibrium: the coarse-graining has fully collapsed the fibre-label information. It is realised by the thermodynamic universal profile  $\sigma^*$  in the limit  $q \rightarrow \infty$  (Remark 5.3).*

**Explicit disclaimer.** Theorem 4.3 is *not* a  $c$ -theorem. At the fibre-erasure equilibrium,  $I_\infty = 0$  does not count any effective degrees of freedom. A  $c$ -theorem analog requires a separate counting functional; see open problem [O-2].

### 4.3 Information decomposition and programme connections

By the chain rule for entropy:

$$H(\sigma(\ell)) = H(\sigma(\ell) \mid c) + I(c; \sigma(\ell)). \quad (5)$$

Here  $I(c; \sigma(\ell))$  is the fibre-erasure functional (Theorem 4.3), monotone non-increasing in  $\ell$ ;  $H(\sigma(\ell) \mid c)$  is the intrinsic entropy of the observable conditioned on the fibre label, and is the natural candidate for a counting monotone [O-2].

*Remark 4.5* (Connection to PTO and axis orthogonality). The temporal monotone  $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$  of [4] lives on the temporal axis  $n$  (Table 1). The present  $I(c; \sigma(\ell))$  lives on the coarse-graining axis  $\ell$ . They measure distinct phenomena on orthogonal axes and should not be conflated.

## 5 Numerical Proxies and Spectral Programme Connections

### 5.1 Inter-sector variance as proxy

**Definition 5.1** (Inter-sector variance). The *inter-sector variance* at depth  $\ell$  is

$$\mathcal{V}(\ell) = \text{Var}_c(\sigma_c(\ell)) = \mathbb{E}_c[\sigma_c(\ell)^2] - (\mathbb{E}_c[\sigma_c(\ell)])^2. \quad (6)$$

$\mathcal{V}(\ell) = 0$  is a necessary condition for  $I(c; \sigma(\ell)) = 0$ , but  $\mathcal{V}$  is *not* guaranteed to be monotone in  $\ell$ . Non-monotone behaviour of  $\mathcal{V}$  would not contradict Theorem 4.3; it would indicate that variance is an imperfect surrogate for  $I(c; \sigma(\ell))$ .

### 5.2 Triplet variance and [H-color]

For  $q \equiv 1 \pmod{3}$ , the *triplet-restricted variance* is

$$\mathcal{V}_{\text{triplet}}(\ell) = \text{Var}(\sigma_c(\ell), \sigma_{\omega c}(\ell), \sigma_{\omega^2 c}(\ell)), \quad (7)$$

averaged over colour triplets.

*Remark 5.2* ([H-color] as a fibre-erasure statement). Hypothesis [H-color] of [8, 7] asserts  $\sigma_c(\ell) = \sigma_{\omega c}(\ell) = \sigma_{\omega^2 c}(\ell)$  for  $q \equiv 1 \pmod{3}$ , which is precisely  $\mathcal{V}_{\text{triplet}}(\ell) = 0$  and, under [H-suff],  $I_{\text{triplet}}(c; \sigma(\ell)) = 0$ . The fibre-erasure theorem provides the information-theoretic framework within which [H-color] is a special case: the colour label becomes unobservable at the effective level. The triplet-restricted variance is already computable from the  $q \in \{61, 151, 211, 307\}$  runs of O32 [7].

### 5.3 Numerical evidence for [H-suff]

Hypothesis [H-suff] was tested numerically on the BFS capacity data from the O25 pipeline [5, 6] for  $q \in \{61, 151, 211\}$ . For each depth step  $\ell \rightarrow \ell + 1$ , the  $k$ -nearest-neighbour regression residuals  $e_c = \sigma_c(\ell + 1) - \hat{f}(\sigma_c(\ell))$  are computed from all  $q - 1$  block-averaged mean trajectories, where  $\hat{f}$  is estimated without reference to  $c$ . The Spearman correlation  $|\rho(e_c, c)|$  is then computed and compared to a permutation null (500 shuffles of  $c$ ), yielding a right-tail  $p$ -value and a standardised  $z$ -score. Under [H-suff], the residuals  $e_c$  should be independent of  $c$ , and  $|\rho|$  should be statistically indistinguishable from zero at every depth.

| $q$ | depths tested | $ \rho _{\max}$ | $z$ -score max | signif. ( $p < 0.05$ ) |
|-----|---------------|-----------------|----------------|------------------------|
| 61  | 10            | 0.169           | 0.78           | 0/10                   |
| 151 | 37            | 0.123           | 1.05           | 0/37                   |
| 211 | 49            | 0.120           | 1.55           | 0/49                   |

Table 2: Spearman test of conditional independence  $\sigma(\ell + 1) \perp c \mid \sigma(\ell)$  on BFS capacity data for  $q \in \{61, 151, 211\}$ .  $|\rho|_{\max}$ : maximum absolute Spearman correlation between residuals and the fibre label over all tested depth steps.  $z$ -score: standardised deviation from the permutation null. Significant: number of depth steps with  $p < 0.05$ ; zero in all cases.

For all three primes and across all depth steps tested,  $|\rho(e_c, c)|$  does not exceed 0.17 and no depth step yields  $p < 0.05$ . The results, shown in Figure 1, are consistent with the permutation null throughout: no systematic  $c$ -dependence in the step-by-step residuals is detected.

These results constitute numerical evidence that the BFS depth-increment dynamics does not depend on the fibre label  $c$  beyond the capacity profile  $\sigma(\ell)$ , consistent with [H-suff].

**Limitations.** The test is limited by the available data: the O25 pipeline [5, 6] stores only the mean BFS capacity profile averaged over  $M$  block samples per fibre label; per-block trajectories would enable a more powerful categorical test. No full O25 run is currently available for  $q = 307$ , though [H-color] at  $q = 307$  has been confirmed directly by O32 [7].

**[H-color] co-admissibility diagnostic.** The leave-triplet-out  $\eta^2$  test (panel C of Figure 1) detects strong within-triplet residual correlation at  $q = 151$  ( $z_{\max} = 3.07$ ) and moderately at  $q = 211$  ( $z_{\max} = 3.25$ ), consistent with the [H-color] co-admissibility established by O32 [7]: triplet members share equal  $\sigma$  profiles, giving equal residuals after leave-triplet-out regression. This is not evidence against [H-suff]; it is an incidental corroboration of [H-color]. The signal increases with better [H-color] quality:  $q = 151$  has the smallest inter-triplet variance ratio ( $R_{\text{var}} = 0.71 \times 10^{-3}$ ) of the tested primes, consistent with the strongest triplet signal.



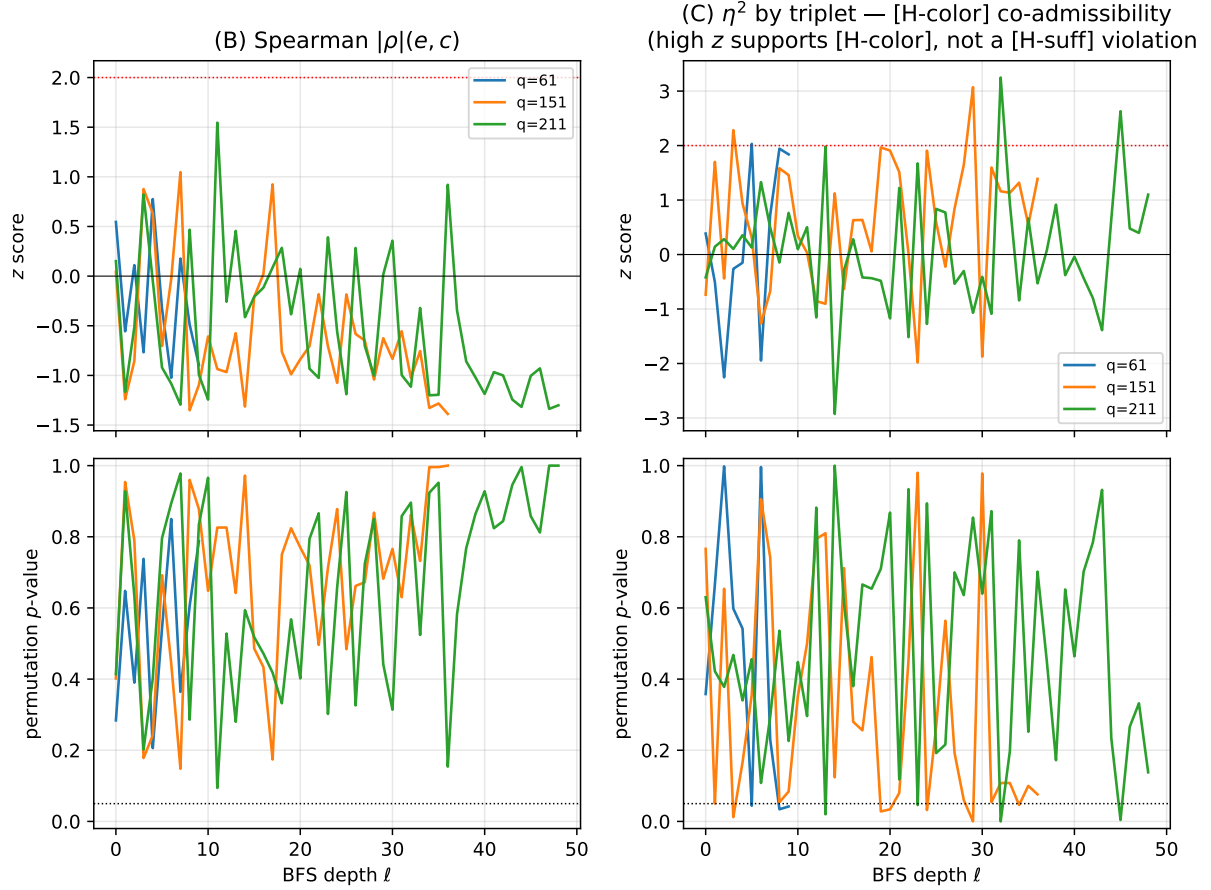


Figure 1: Numerical test of [H-suff] for  $q \in \{61, 151, 211\}$ . **Left panels (B):** Spearman  $|\rho|(e, c)$  test ( $z$ -scores and  $p$ -values vs BFS depth  $\ell$ ). All  $z$ -scores lie below the  $z = 2$  threshold (dashed red); no depth is significant. **Right panels (C):** leave-triplet-out  $\eta^2$  test grouped by SU(3) colour triplet for  $q \equiv 1 \pmod{3}$ ; this measures [H-color] co-admissibility quality and does not test [H-suff] directly (see text).

## 5.4 Axis disambiguation: the fixed point is thermodynamic

The question of whether the fibre-erasure fixed point  $\sigma^*$  — the configuration where  $I(c; \sigma(\ell)) = 0$  and fibre-label information is fully collapsed — is approached as a limit in the coarse-graining depth  $\ell$  at fixed  $q$  (a Wilsonian fixed point of the semigroup  $\{G_{\ell' \leftarrow \ell}\}$ ) or only in the thermodynamic limit  $q \rightarrow \infty$  was investigated numerically on the BFS capacity data for  $q \in \{61, 151, 211\}$ . Two complementary diagnostics were computed.

**Coefficient of variation.** The scale-free inter-sector spread

$$\text{CV}(\ell) = \frac{\text{std}_c(\sigma_c(\ell))}{\text{mean}_c(\sigma_c(\ell))} \quad (8)$$

equals  $\text{std}(A_c)/\text{mean}(A_c)$ , a depth-independent constant, if and only if profiles differ purely by amplitude:  $\sigma_c(\ell) = A_c \sigma^*(\ell)$ . Decrease of  $\text{CV}(\ell)$  to zero would indicate convergence to a common profile, i.e. a Wilsonian fixed point at fixed  $q$ . The data show  $\text{CV}(\ell)$  starting small at the fitting-window entry (0.045–0.079 across  $q \in \{61, 151, 211\}$ ) and growing throughout the pre-saturation regime at all three primes (Figure 2, left panels); no plateau or decrease toward zero is observed. This rules out Wilsonian convergence at fixed  $q$ .

**Amplitude-ratio test.** For 20 random pairs  $(c, c')$ , the coefficient of variation of the ratio  $\sigma_c(\ell)/\sigma_{c'}(\ell)$  over depth measures shape differences beyond amplitude scaling. Table 3 shows this quantity decreasing with  $q$ . The trend from  $q = 61$  (0.156) to  $q \in \{151, 211\}$  (0.065–0.062) is

| $q$ | ratio-CV mean | ratio-CV std |
|-----|---------------|--------------|
| 61  | 0.156         | 0.069        |
| 151 | 0.065         | 0.021        |
| 211 | 0.062         | 0.025        |

Table 3: Amplitude-ratio CV over 20 random pairs  $(c, c')$ : coefficient of variation of  $\sigma_c(\ell)/\sigma_{c'}(\ell)$  over depth in the pre-saturation regime. The decrease with  $q$  is consistent with  $O(q^{-1/2})$  shape corrections from U1 universality [5, 6].

consistent with  $O(q^{-1/2})$  corrections from U1 universality [5, 6]: at larger  $q$ , profiles approach a common shape.

**Cross- $q$  mean profile comparison.** After normalising each mean profile by its value at the fitting-window start  $\ell_0 = n_0$ , the profiles for  $q = 151$  and  $q = 211$  are nearly identical, while  $q = 61$  decays more rapidly (Figure 2, right panel). No collapse onto a single  $q$ -universal curve is observed at finite  $q$ .

These results establish that  $\sigma^*$  is a thermodynamic limit object: it is approached as  $q \rightarrow \infty$  with  $O(q^{-1/2})$  amplitude corrections from U1 universality, not as a Wilsonian  $\ell$ -limit at fixed  $q$ . At finite  $q$ , approximate fixed points  $\sigma^*(q)$  exist with  $c$ -dependent amplitude corrections  $A_c$  that vanish as  $q \rightarrow \infty$ ; the universal fixed point requires the continuum limit.

*Remark 5.3* (Residual fibre information at finite  $q$ ; terminology). Theorem 4.3 and Corollary 4.4 hold for any fixed  $q$ : their conclusion is that  $I(c; \sigma(\ell))$  is non-increasing; they do not require the equilibrium condition to be reached. At finite  $q$ , a residual  $I(c; \sigma(\ell)) > 0$  persists at all depths, reflecting the  $c$ -dependent amplitude corrections  $A_c$ .

A terminological precision is warranted. The *fibre-erasure fixed point* of Corollary 4.4 is the equilibrium condition  $I(c; \sigma) = 0$  on the information functional — not a fixed point of the semigroup  $\{G_{\ell' \leftarrow \ell}\}$ , which has no non-trivial fixed point at finite  $q$  (all profiles eventually reach  $\sigma_c(\ell) \rightarrow 0$  by BFS saturation). The universal profile  $\sigma^*$  realising this equilibrium condition is

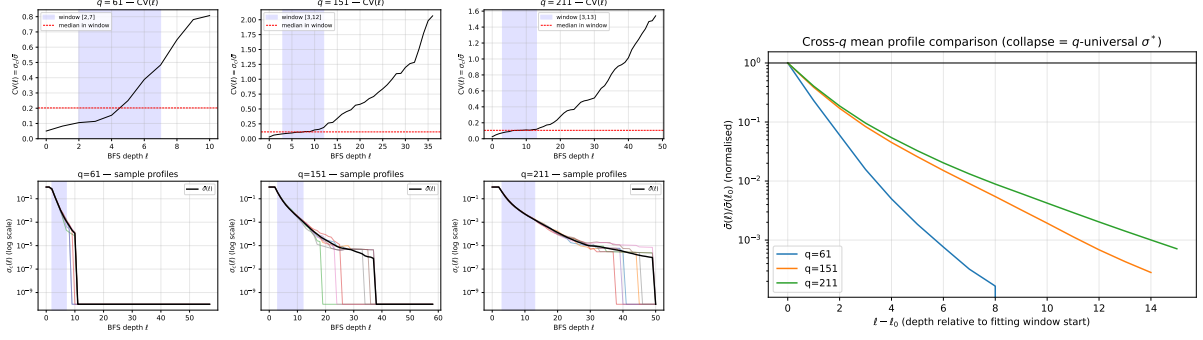


Figure 2: **Left:**  $CV(\ell)$  (top row) and sample profiles on a log scale (bottom row) for  $q \in \{61, 151, 211\}$ . Blue band: fitting window  $[n_0, n_1]$ .  $CV(\ell)$  grows monotonically throughout the pre-saturation regime; the sharp rise beyond the window is a saturation artefact (different  $c$  values reach  $\sigma_c \rightarrow 0$  at different depths). No plateau or decrease toward zero is observed. **Right:** cross- $q$  mean profiles normalised at  $\ell_0 = n_0$ . The  $q = 151$  and  $q = 211$  profiles are nearly identical;  $q = 61$  decays faster. The absence of collapse confirms that  $\sigma^*$  requires  $q \rightarrow \infty$ .

therefore better understood as the *thermodynamic universal profile*:  $\sigma^* = \lim_{q \rightarrow \infty} \sigma^*(q)$ , where  $\sigma^*(q)$  are approximate equilibria at finite  $q$  with  $I(c; \sigma^*(q)) > 0$ , approaching exact equilibrium only in the projective continuum limit  $q \rightarrow \infty$  (Conjecture 4.7 of [2]).

## 5.5 Numerical evidence for [O-2]: conditional entropy monotonicity

The decomposition (5) identifies  $H(\sigma(\ell) \mid c)$  as a candidate counting monotone (open problem [O-2]). A numerical proxy is available from the O25 pipeline extended with squared-profile accumulators: for each pair  $(c, q - c)$  and BFS depth  $\ell$ , the Bessel-corrected intra-block variance

$$\text{Var}_{\text{intra}}(\ell, c) = \frac{M}{M-1} \left( \mathbb{E}_{\text{block}}[\sigma_c(\ell)^2] - \mathbb{E}_{\text{block}}[\sigma_c(\ell)]^2 \right) \quad (9)$$

estimates the spread of  $\sigma_c(\ell)$  across the  $M$  independent block samples for fixed fibre label  $c$ . Under a Gaussian approximation,  $H(\sigma(\ell) \mid c) \approx \text{mean}_c \left[ \frac{1}{2} \log(2\pi e \text{Var}_{\text{intra}}(\ell, c)) \right]$ , and its monotonicity reduces to that of the proxy

$$h_{\text{intra}}(\ell) = \text{mean}_c \left[ \frac{1}{2} \log \text{Var}_{\text{intra}}(\ell, c) \right]. \quad (10)$$

**Result for  $q = 61$ .** On the BFS capacity data for  $q = 61$  (30 conjugate pairs,  $M = 20$  block samples, fitting window  $[2, 7]$ ),  $h_{\text{intra}}(\ell)$  is perfectly monotone non-increasing over the pre-saturation regime: zero violations over five consecutive depth increments, linear-trend  $t$ -statistic  $-25.5$ , and a reduction of the intra-block standard deviation by a factor 0.024 (approximately  $41\times$ ) over the window. Figure 3 shows  $h_{\text{intra}}(\ell)$ , the between- $c$  entropy proxy  $h_{\text{inter}}(\ell)$ , and the normalised spread  $\text{CV}_{\text{intra}}(\ell)$ .

The normalised spread  $\text{CV}_{\text{intra}}(\ell) = \text{mean}_c [\text{std}_{\text{intra}} / \bar{\sigma}_c]$  increases over the same range ( $0.46 \rightarrow 2.05$ ). This reflects the staggered-saturation artefact identified in Section 5.4: at late depths, some of the  $M = 20$  blocks reach  $\sigma_c = 0$  while others have not, inflating relative within- $c$  variance. It is not a violation of [O-2]; the absolute proxy  $h_{\text{intra}}$  is the correct test statistic.

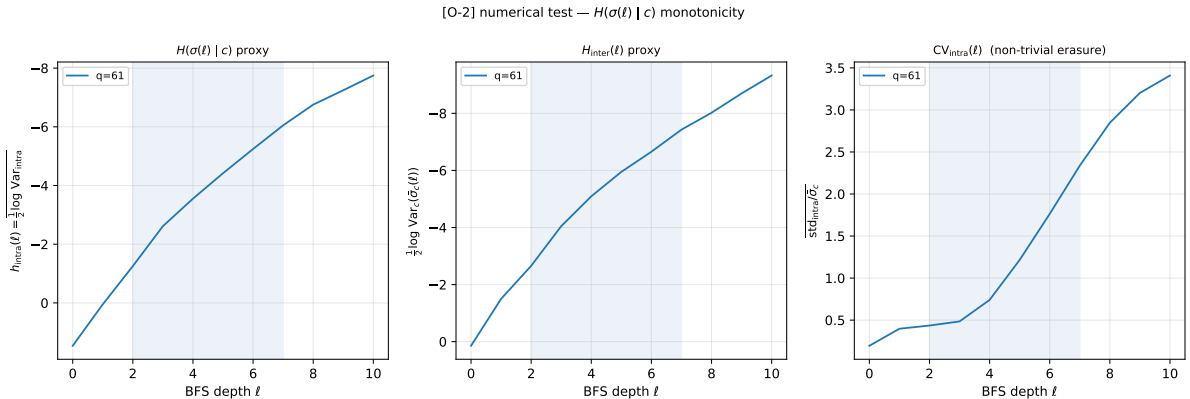


Figure 3: Preliminary numerical test of [O-2] for  $q = 61$ . **Left:**  $h_{\text{intra}}(\ell)$ , proxy for  $H(\sigma(\ell) \mid c)$ ; perfectly monotone non-increasing over the fitting window (blue band), zero violations,  $t = -25.5$ . **Centre:**  $h_{\text{inter}}(\ell)$ , between- $c$  entropy proxy. **Right:**  $\text{CV}_{\text{intra}}(\ell)$ ; the increase at late depths is the staggered-saturation artefact (Section 5.4), not a violation of [O-2].

These results constitute preliminary numerical evidence that  $H(\sigma(\ell) \mid c)$  is monotone non-increasing, consistent with [O-2].

**Status of the Gaussian proxy.** The quantity  $h_{\text{intra}}(\ell)$  is not an exact Shannon entropy unless the within- $c$  distribution of  $\sigma_c(\ell)$  is Gaussian. It should be understood as a variance-based entropy proxy. Its monotonicity is nevertheless meaningful: any unimodal within- $c$  distribution whose scale concentrates with  $\ell$  has decreasing entropy under standard scale-sensitive estimators. Thus the present test probes concentration of the intrinsic observable spread, not a full non-parametric entropy estimate.

**Counting requirement.** A genuine counting monotone requires more than monotonicity alone. One must also show that  $H(\sigma(\ell) \mid c)$  approaches a non-zero, stable, and interpretable value at the approximate equilibrium  $\sigma^*(q)$ , with a controlled  $q$ -dependence. If this limiting value vanished identically, the functional would not count effective degrees of freedom in the sense required by a  $c$ -theorem analog.

**Role of the multi- $q$  extension.** The planned  $q \in \{101, 151, 211\}$  tests are not merely robustness checks. They should determine whether the decay rate of  $h_{\text{intra}}(\ell)$  is coherent across  $q$ , and whether it scales compatibly with the decay of the fibre-erasure functional  $I(c; \sigma(\ell))$ . If the two terms of decomposition (5) decay with compatible exponents, the decomposition becomes an operational structural statement rather than a purely formal identity.

## 6 Open Problems

### [O-1] Analytical proof of [H-suff].

Establish that the BFS depth-increment map  $G_{\ell+1 \leftarrow \ell}$  is a Markov kernel on  $\sigma$ : the transition  $p(\sigma(\ell+1) \mid \sigma(\ell), c)$  is independent of  $c$ . Numerical evidence is available for  $q \in \{61, 151, 211\}$  via the Spearman residual test of Section 5.3; the analytical proof is the remaining open task. The structural route would show that the BFS recursion depends on the Heisenberg group geometry — shared across all fibre labels up to conjugation — and the Born-Infeld admissibility constraint, not on the specific character of block  $c$ .

### [O-1b] Proof of the semigroup hypothesis [H-sg].

Show that the coarse-graining channels compose as  $G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1}$  for all  $\ell_1 < \ell_2 < \ell_3$ . Without [H-sg], the data-processing inequality yields step-by-step monotonicity (from each application of [H-suff]) but not the uniform statement of Theorem 4.3 for all  $\ell' > \ell$ . Neither a numerical test nor an analytical proof is currently available.

### [O-2] $c$ -theorem analog via $H(\sigma(\ell))$ .

The decomposition (5) identifies  $H(\sigma(\ell) \mid c) = H(\sigma(\ell)) - I(c; \sigma(\ell))$  as a candidate counting monotone. Establish its monotonicity in  $\ell$  and verify that its fixed-point value counts effective degrees of freedom. This would upgrade the fibre-erasure theorem to a genuine  $c$ -theorem analog. Preliminary numerical evidence for the monotonicity is available for  $q = 61$  via the intra-block variance proxy (Section 5.5); analytical proof and cross- $q$  confirmation ( $q \in \{101, 151, 211\}$ ) remain open.

### [O-3] Temporal causal structure and emergent light cone.

Define the temporal mutual information  $I_{\text{temp}}(n, k) = I(U_n; U_{n+k})$  and the transfer entropy  $T(U_n \rightarrow U_{n+k}) = I(U_{n+k}; U_n \mid U_{n-1}, \dots, U_0)$ . Their decay in  $k$  would yield an operational definition of the emergent forward light cone on the temporal axis  $n$ , orthogonal to the present paper.

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